Mumbai University Bachelor's of Computer Engineering(Syallbus)

Semester III

Course Code	Course Name	Module name
2113111	Mathematics for Computer Engineering	1. Linear Algebra (Theory of Matrices) 2. Linear and Non-Linear Programming Problems 3. Modular Arithmetic 4. Fourier Series 5. Statistical Techniques 6. Probability
2113112	Discrete Structures and Graph Theory	1. Crisp Set Theory and Logic 2. Mathematical Relations 3. Functions 4. Counting 5. Algebraic Structures 6. Graph Theory
2113113	Analysis of Algorithm	1. Introduction 2. Divide and Conquer Approach 3. Greedy Method Approach 4. Dynamic Programming Approach 5. Backtracking and Branch and bound 6. String Matching Algorithms
2113114	Computer organization & Architecture	1. Computer Fundamentals 2. ALU Operations 3. Processor Organization and Control Unit Design 4. Memory Systems Organization 5. I/O Organization 6. Parallel Processing
2113311	Open Elective	
2113611	Full Stack Java Programming	1. Introduction to OOP in Java 2. Inheritance & Exception Handling 3. Client and Server Side Programming 4. Fundamentals of Java Script 5. Web Programming using React 6. Applications of Spring Framework
2993511	Entrepreneurship Development	1. Introduction to Entrepreneurship 2. Business Idea Generation & Validation 3. Business Planning & Strategy 4. Funding and Legal Framework 5. Marketing & Digital Presence

		6. Business Pitching & Prototype Development
2993512	Environmental Science for Engineers	1. The Multidisciplinary Nature of Environmental Studies 2. Natural Resources 3. Ecosystems 4. Biodiversity and its Conservation 5. Environmental Pollution Definition 6. Social Issues and Environment

Semester IV

Course Code	Course Name	Module Name
2114111	Computational Theory	1. Basics Concepts and Regular Languages 2. Finite Automata 3. Regular and Context Free Grammars 4. Pushdown Automata (PDA) 5. Turing Machine (TM) 6. Decidability and Computability
2114112	Database Management System	1. Introduction to Database and Data Modeling 2. Relational Model and Relational Algebra 3. Structured Query Language (SQL) 4. Database Normalization 5. Transaction Management and Concurrency Control 6. Introduction to Modern databases
2114113	Operating System	1. Fundamentals of Operating System 2. Process Management 3. Process Synchronization and Deadlock Management 4. Memory Management 5. File and IO Management 6. Special-purpose Operating Systems
MDC401	Multidisciplinary minor	
2114311	Open Elective	
2994511	Business Model Development	1. Introduction to Entrepreneurship 2. Entrepreneurship Development 3. Start-up financing 4. Intellectual Property Rights (IPR) 5. Business Model Development 6. Digital Business Management

2994512	Design Thinking	1. Introduction to Design Thinking 2. Empathy 3. Define 4. Ideate 5. Prototype 6. Test